3. Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

CODING:

```
import math

data = [
  ['Sunny', 'Hot', 'No'],
  ['Sunny', 'Cool', 'Yes'],
  ['Rainy', 'Hot', 'No'],
  ['Rainy', 'Cool', 'Yes'],
  ['Overcast', 'Hot', 'Yes'],
  ['Overcast', 'Cool', 'Yes']
]

def entropy(rows):
  counts = {}
  for row in rows:
    counts[row[-1]] = counts.get(row[-1], 0) + 1

  result = 0
  for count in counts.values():
    p = count / len(rows)
    result -= p * math.log2(p)
  return result

def build_tree(rows, features):
  labels = [row[-1] for row in rows]
```

```
    if len(set(labels)) == 1:
        return labels[0]

    if not features:
        return max(set(labels), key=labels.count)

    best = max(
        range(len(features)),
        key=lambda i: entropy(rows) - sum(
            len([r for r in rows if r[i] == v]) / len(rows)
            * entropy([r for r in rows if r[i] == v])
            for v in set(r[i] for r in rows)
        )
    )

    feature = features[best]
    tree = {feature: {}}

    for value in set(row[best] for row in rows):
        subset = [r[:best] + r[best+1:] for r in rows if r[best] == value]
        tree[feature][value] = build_tree(
            subset, features[:best] + features[best+1:]
        )

    return tree

features = ['Weather', 'Temperature']
```

tree = build_tree(data, features)

print("Decision Tree:")

print(tree)

OUTPUT:

